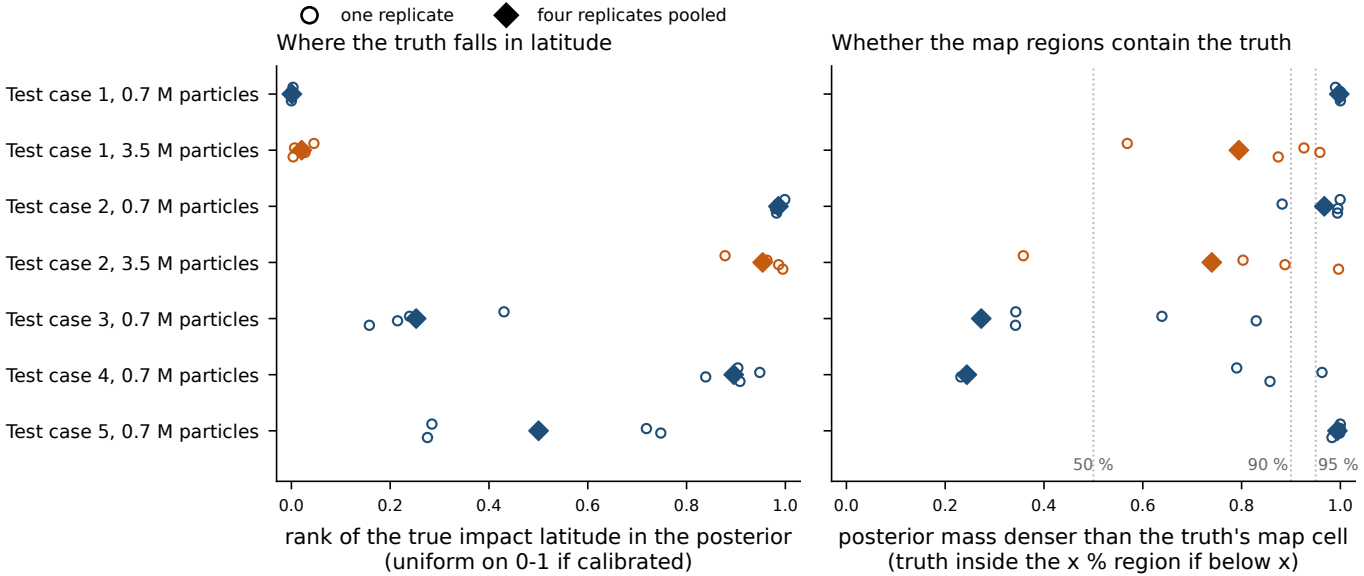
Monte Carlo noise of the calibration statistics at two sample sizes



Note (simplified English): for each test case we show where the known truth falls in each of the four replicate posteriors (open circles) and in the pooled posterior (diamond). When the circles of one case spread widely, the Monte Carlo noise is too large for a calibration test. With 3.5 million particles per replicate the posterior starts to contain the truth in test cases 1 and 2, and the spread becomes smaller. The 00:19 data are held out; core evidence only. Physics of the end of flight is provisional.

Technical: statistics per protocol 5 (threads/work-orders/e2e-calibration-protocol-2026-10-19.md): normalised rank u = P_post(lat < lat_true); map level = posterior mass in equal-area cells (0.25° x 769 km²) denser than the truth's cell. Option D0 (held out), log-on cause mixture q = 1/3, survival factor; waypoints stratum only; 4 replicates; 0.7 M = 1/10 and 3.5 M = 1/2 of the run J production particles per seed; 10^4 and 2×10^4 hand-off rows, 64 descents each. Truths from the fuel-only engine prior with the data event BTO(00:11) 17,540-18,540 µs, BFO(00:11) 202-302 Hz. Five cases cannot test coverage: smoke only.